A 16-Bit-Native Entropy Coding Pipeline for Lossless Medical Image Compression

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***Abstract*—Lossless compression remains important in medical imaging because archival storage, transmission, and diagnostic workflows often require exact reconstruction of high-bit-depth pixel data. Many medical images are stored at 10–16 bits per sample, yet most practical compression software remains optimized for byte-oriented processing or small symbol alphabets. This paper studies a lossless compression pipeline designed to operate natively on predicted 16-bit residuals.**

We present MIC, a codec consisting of three stages: spatial prediction, 16-bit run-length encoding, and large-alphabet table-based asymmetric numeral systems (ANS) coding. In contrast to codecs that split 16-bit pixels into byte pairs before entropy coding, MIC treats each predicted residual as a single 16-bit symbol, allowing the entropy model to capture correlations between the high and low byte directly. The implementation extends Finite State Entropy (FSE), a table-driven ANS variant, to support large active symbol sets arising in medical image residual streams and includes a multi-state interleaved decoder that increases instruction-level parallelism (ILP) during decompression. The design is motivated by the observation that, after simple spatial prediction, many medical images produce residual distributions sharply concentrated near zero with frequently repeated 16-bit residual values.

Evaluation on 21 de-identified DICOM images spanning nine modalities shows the 16-bit-native pipeline improves compression ratio by 5–22% (geometric mean +14%) relative to a delta + Zstandard baseline, with a geometric-mean ratio of 3.12×—within 1% of High-Throughput JPEG 2000 (HTJ2K, 3.15×)—while being structurally simpler. Across grayscale images, ratios range from 1.7× to 8.9×. A four-state interleaved decoder yields a geometric-mean +68% isolated FSE speedup on ARM64; single-threaded MIC exceeds HTJ2K decompression speed on the majority of tested images, and strip-parallel configurations exceed it on all 21 images on both ARM64 and AMD64. A compact JavaScript implementation (∼20 KB, zero dependencies) and a Go WebAssembly build confirm that the pipeline is lightweight enough for client-side in-browser decoding, suggesting 16-bit-native entropy coding is a practical design point when throughput, simplicity, and deployment portability matter.

***Index Terms*—medical image compression, lossless compression, entropy coding, asymmetric numeral systems (ANS), finite state entropy (FSE), DICOM, high-bit-depth imaging.**

I. Introduction

MEDICAL imaging systems routinely generate large volumes of high-bit-depth data. Modalities such as computed tomography (CT), magnetic resonance imaging (MR), digital radiography (computed radiography, CR; plain X-ray, XR), mammography, and tomosynthesis commonly store pixel values at 10, 12, or 16 bits per sample. A single digital breast tomosynthesis (DBT) acquisition view produces 60 or more reconstructed slices (for example, 69 frames at 2457×1890 pixels, 10-bit depth), totalling over 600 MB of raw pixel data per view. A complete bilateral four-view screening study exceeds 2 GB uncompressed. Efficient lossless compression is essential for archival storage, network transmission, and real-time rendering in clinical Picture Archiving and Communication Systems (PACS) and diagnostic viewers.

DICOM [1] supports several established lossless transfer syntaxes, including JPEG 2000, HTJ2K, JPEG-LS (a lossless/near-lossless predictive standard), and Run-Length Encoding (RLE) Lossless. These methods offer different tradeoffs among compression ratio, decoding speed, and implementation complexity [24], [25]. JPEG 2000 and HTJ2K provide strong compression performance but rely on transform and block-coding pipelines that are comparatively complex. JPEG-LS uses predictive coding and often yields strong lossless ratios, but its throughput characteristics remain application-dependent. RLE Lossless is simple but typically provides limited compression because it operates on byte-oriented runs and does not include an effective decorrelation stage.

This work studies a different design point for lossless medical image compression: direct coding of **predicted 16-bit residual streams**. Throughout this paper, a *residual stream* denotes the sequence of per-pixel prediction errors $r_{i,j} = x_{i,j} - \hat{x}_{i,j}$ produced by a spatial predictor, kept at the native 16-bit precision of the source pixels rather than re-serialised into bytes. As a concrete example: if the predicted pixel value is 1020 and the true pixel is 1023, the residual is $+3$. Smooth regions therefore produce many residuals near zero, and these near-zero residuals recur frequently—properties that motivate a 16-bit-native run-length and entropy coding stage.

The central observation is that the dominant entropy coders in modern compression systems, including Huffman coding, arithmetic coding, Lempel-Ziv 1977 (LZ77), and FSE/ANS as used in Zstandard, were designed primarily for 8-bit byte streams. Medical images break this assumption: DICOM stores pixels at 10–16 bits per sample, yielding symbol alphabets of 1,024 to 65,536 entries. Even after spatial delta prediction, the residual alphabet for a 16-bit CT image can span tens of thousands of distinct values. An entropy coder that assumes an 8-bit alphabet must either re-encode 16-bit symbols as byte pairs (losing inter-byte correlation and doubling the number of coding steps) or truncate the alphabet at 256 and fall back to raw storage for rare symbols. This mismatch is structurally costly: for example, small signed residuals such as $-2, -1, 0, +1$ have highly predictable high bytes, so encoding the high and low bytes independently forces the coder to spend bits on information already implied by the other byte. On our test

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Code: https://github.com/pappuks/medical-image-codec.

images, the mutual information $I(X_H; X_L)$ between the high and low bytes of delta residuals ranges from 0.3 bits/symbol (MR) to 1.1 bits/symbol (MG3), corresponding to 8–22% of total entropy and explaining MIC's 5–22% empirical advantage over Delta + Zstandard.

Based on this observation, we present MIC, a lossless codec that combines:

1) a simple spatial predictor,
2) a 16-bit-native run-length representation of the residual stream, and
3) an extended FSE/tANS (table-based ANS) entropy coder supporting large active symbol alphabets.

The codec also includes multi-state interleaved entropy decoding to improve decompression throughput by increasing ILP.

The paper is motivated by three practical questions:

1) Can a 16-bit-native entropy coding pipeline improve compression efficiency over byte-oriented baselines on medical residual streams?
2) Can entropy-decoder organization be modified to improve throughput without changing compressed size?
3) Can such a codec remain compact enough for lightweight deployment, including client-side decoding?

A. Contributions

The main contributions of this paper are as follows.

1) **A 16-bit-native lossless coding pipeline for medical images.** We present a Delta + 16-bit RLE + extended FSE pipeline for grayscale medical image compression that consistently outperforms Delta + Zstandard on all 21 tested images, with the improvement attributable to the structural cost of byte-splitting 16-bit residuals.
2) **An extended table-based entropy coder for large residual alphabets.** The proposed implementation supports up to 65,535 distinct symbols. One 16-bit code value ($2^{16}-1 = 65535$) is reserved as the overflow delimiter for full-range images, leaving up to 65,535 codes available for entropy coding. Tables are dynamically sized to the observed symbol range, reducing working-set size from 512 KB to approximately 2 KB for 8-bit images.
3) **A multi-state interleaved entropy decoder.** We implement and evaluate two-state and four-state FSE decoding to exploit ILP, achieving a geometric-mean +68% isolated FSE speedup on ARM64 (+43% on AMD64) with no compression ratio penalty.
4) **An empirical evaluation against standard codecs and a general-purpose baseline.** We compare MIC with HTJ2K (OpenJPH) [29], JPEG-LS (CharLS) [30], and Delta + Zstandard on a dataset of 21 public de-identified DICOM images spanning 10 modalities, using fair in-process benchmarking via CGO (Go's C foreign-function interface) bindings.
5) **A compact JavaScript/WebAssembly (WASM) decoder implementation.** We provide a ∼20 KB pure JavaScript (JS) decoder and a Go WebAssembly build to explore deployment feasibility for client-side viewing.

B. Scope

This paper is scoped to **lossless single-frame grayscale medical image compression**. Multi-frame and RGB extensions are outside the scope of this work. The present work does not address lossy coding, progressive decoding, or region-of-interest coding.

C. Organization

Section II reviews related work and positions MIC relative to existing methods. Section III describes the proposed coding pipeline. Section IV presents the multi-state entropy decoder. Section V describes experimental methodology. Section VI reports compression and speed results. Section VII discusses limitations and implications. Section VIII concludes.

II. Related Work

A. Lossless Compression in Medical Imaging

JPEG 2000 [2] employs the reversible 5/3 wavelet transform [20], [23] with Embedded Block Coding with Optimized Truncation (EBCOT), achieving excellent ratios at significant decompression overhead. HTJ2K [3], [27] replaces EBCOT with a faster block coder, substantially improving decode speed. JPEG-LS [4], [5] uses context-adaptive Median Edge Detector (MED) prediction with Golomb-Rice coding; it is an important lossless baseline as a standardized predictive codec widely deployed in medical PACS. RLE Lossless (DICOM TS 1.2.840.10008.1.2.5) applies byte-level PackBits without decorrelation, yielding poor ratios (typically $<1.5\times$). Higher-complexity approaches such as the Context-based Adaptive Lossless Image Codec (CALIC) [18] and the Free Lossless Image Format (FLIF) [19] achieve strong ratios on natural images but lack DICOM standardization and browser decoders.

These codecs are the primary baselines for this work. The goal is not to supersede them but to investigate a simpler residual-domain design specifically targeting high-bit-depth images with strong throughput and deployment portability.

B. Entropy Coding and the Byte Assumption

Asymmetric numeral systems (ANS) [7], [8] replace arithmetic coding's interval subdivision with integer state transitions. Finite State Entropy (FSE) [9] is a table-driven tANS implementation with $\mathcal{O}(1)$ per-symbol operations, popularized by Zstandard [10]. Recent work has formalized ANS efficiency bounds [12] and statistical interpretations [13]. In practice, Huffman coding becomes impractical above ∼4,096 symbols and FSE in Zstandard is capped at 4,096; modern formats such as JPEG XL [6] likewise target byte streams.

Information-theoretic cost of byte-splitting. For a 16-bit symbol X decomposed into high byte X_H and low byte X_L, a byte-level coder achieves at best $H(X_H) + H(X_L) = H(X) + I(X_H; X_L)$. After delta prediction, when a residual is small (for example, ±30), X_H is nearly deterministic given X_L, but it must still be encoded independently. On our test images, $I(X_H; X_L)$ ranges from 0.3 bits/symbol (MR) to 1.1 bits/symbol (MG3), corresponding to 8–22% of total entropy and closely matching MIC's empirical advantage over Delta + Zstandard.

C. ANS and Interleaved Decoding

Prior work has shown that interleaving or multi-stream organization can improve ANS decoder throughput. Giesen [14], [15] demonstrated the value of multi-stream range ANS (rANS); Lin *et al.* [16] extended parallel rANS to decoder-adaptive scalability; Weissenberger and Schmidt [17] showed massively parallel ANS decoding on GPUs. The present work does not claim novelty in ANS itself or in interleaving as a general concept. Rather, the contribution lies in applying these ideas in a 16-bit-native medical image codec, together with a specific large-alphabet implementation and throughput-oriented software design.

D. Positioning of This Work

Table I distinguishes MIC from prior work across five dimensions relevant to medical image compression.

TABLE I: Feature comparison of MIC with related codecs and entropy coding methods.

Feature	JPEG-LS	HTJ2K	Zstd	rANS	MIC
Lossless medical	Yes	Yes	General	General	Yes
Native 16-bit	No	No	No	Configurable	Yes
Multi-state ILP	No	No	No	Yes	Yes
Browser decoder	No	No	Partial	No	Yes
Source lines	5k	20k	N/A	N/A	1.1k

III. Proposed Method

MIC compresses grayscale images using three sequential stages: spatial prediction, 16-bit run-length encoding, and large-alphabet table-based entropy coding. Each stage is simple enough to decode in a single sequential pass with minimal branching, as illustrated in Fig. 1.

A. Spatial Prediction

Let $x_{i,j}$ denote the pixel at row i, column j. For interior pixels, MIC predicts the current pixel from the average of the top and left neighbors:

$$\hat{x}_{i,j} = \left\lfloor \frac{x_{i-1,j} + x_{i,j-1}}{2} \right\rfloor \tag{1}$$

where the division is integer division rounded down (matching the decoder exactly), and all arithmetic is performed in unsigned integers. The residual is formed as:

$$r_{i,j} = x_{i,j} - \hat{x}_{i,j}. \tag{2}$$

Boundary pixels use only the available neighbor, and the first pixel is stored directly. This predictor was selected because it is computationally simple, branch-light in decoding, and effective on the current dataset.

Predictor comparison. We implemented and compared four predictors through the full RLE + FSE pipeline on the 21-image dataset: left-only (left neighbor, no top), average (MIC default), Paeth (PNG predictor [26]), and MED (JPEG-LS [4]). Table II summarizes geometric means and relative gains.

MED decompression is 1.5–2× slower due to the three-way conditional branch and diagonal neighbor dependency that prevents the branch-free interior loop used by the average predictor. MED yields the best geomean ratio (+1.6% over Avg) but at significant speed cost; the average predictor is retained as the default for its consistent per-modality performance and branch-free decompression path.

TABLE II: Predictor ablation summary: geometric means and win counts across all 21 images.

Predictor	Geo. Mean	Wins	Avg→X
Left-only	3.38×	3/21	−2.3%
Average (MIC)	3.46×	13/21	baseline
Paeth	3.48×	2/21	+0.5%
MED (JPEG-LS)	3.52×	16/21	+1.6%

B. Effective Bit Depth and Overflow Coding

For 16-bit images, residuals can span $[-65535, +65535]$. Rather than widening the main residual stream to 32 bits, MIC uses an overflow delimiter scheme derived from the effective image bit depth:

$$d = \lceil \log_2(\max(x) + 1) \rceil. \tag{3}$$

Residuals within an in-range interval are mapped directly to 16-bit codes. Residuals outside that interval are encoded using a delimiter followed by the raw pixel value.

Encoding procedure:

1) Compute `deltaThreshold` $= 2^{d-1} - 1$.
2) Compute delimiter $D = 2^d - 1$.
3) For each residual $r_{i,j}$: if $|r_{i,j}| <$ `deltaThreshold`, emit `uint16(deltaThreshold +` $r_{i,j}$`);` otherwise emit delimiter D, then emit raw pixel $x_{i,j}$.

The threshold condition is strict ($<$, not $\leq$), so residuals with $|r| =$ `deltaThreshold` take the overflow path. Residual zero maps to code value `deltaThreshold` exactly. Table III illustrates the mapping for a representative bit depth.

TABLE III: Overflow coding example for $d = 12$ bits (`deltaThreshold`$= 2047$, $D = 4095$).

Residual r	Emitted code(s)	Notes
-2	2045	in-range
-1	2046	in-range
0	2047	in-range (= threshold)
$+1$	2048	in-range
$+2$	2049	in-range
± 2047	4095, raw pixel	overflow

Non-collision proof. In-range residuals satisfy $|r| \leq$ `deltaThreshold` $- 1$, so stored values range from 1 to $2^d - 3$. The delimiter is $D = 2^d - 1$. Since $2^d - 3 < 2^d - 1$, the stored value of a direct residual can never equal D; the two ranges are disjoint.

Decoding procedure: Read the next uint16 value c. If $c \neq D$, recover the residual as $r =$ `int16(`$c-$ `deltaThreshold)` and reconstruct $x_{i,j} = \hat{x}_{i,j} + r$. If $c = D$, the raw pixel follows: read the next uint16 as $x_{i,j}$ directly.

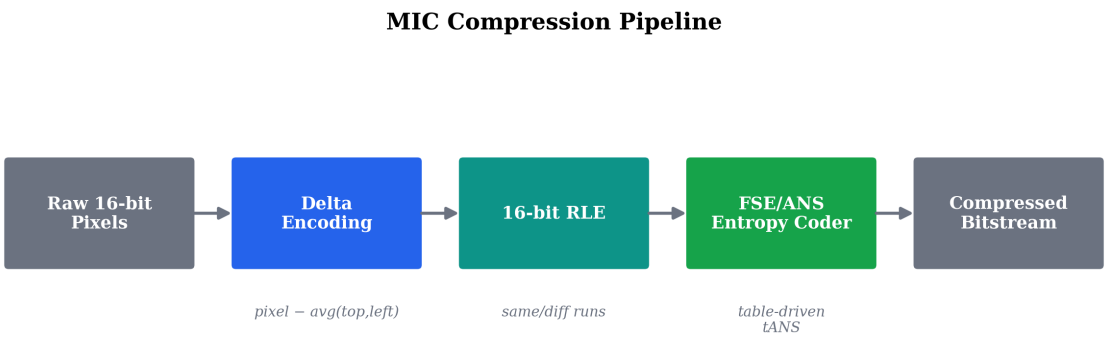

Fig. 1: MIC compression pipeline. Raw 16-bit pixels are delta-encoded (avg of top and left neighbors), run-length encoded (same/diff runs), and entropy-coded with a 16-bit-native FSE/tANS coder.

C. 16-Bit Run-Length Encoding

The predicted residual stream is converted into an alternating sequence of:

- **Same runs**: a count and a repeated 16-bit value;
- **Diff runs**: a count followed by a sequence of distinct 16-bit values.

A same run is emitted only when at least three consecutive identical values are observed.

Midpoint protocol. Run headers are stored as uint16 values. A midpoint constant `midCount`$= 32767$ partitions the header space: a count $c \leq$ `midCount` signals a same run of length c; a count $c >$ `midCount` signals a diff run of length $c-$ `midCount`. The maximum same-run or diff-run length is therefore 32767 symbols; longer sequences are split into multiple headers transparently. The decoder distinguishes the two run types solely by comparing the header value against `midCount`.

Encoding example. For the residual code sequence $[5, 5, 5, 5, 9, 10, 11, 11, 11]$, the RLE output is: same($c = 4$, value=5), diff($c = 2$, values=$9, 10$), same($c = 3$, value=11). The encoded stream contains five uint16 words: $[4,\ 5,\ 32769,\ 9,\ 10,\ 3,\ 11]$, where $32769 =$ `midCount` $+ 2$ signals a diff run of length 2.

Why 16-bit RLE matters. A run of 1,000 identical 16-bit residuals $v =$ `0x0103` is encoded by MIC as two uint16 words (a count and the value), regardless of v's byte pattern. A byte-oriented RLE sees the interleaved stream `0x03,0x01,0x03,0x01,...`: no single byte repeats 1,000 times, so it must emit two interleaved runs or fall back to literals.

Same-run fast path. Same-value runs (often >80% of all runs on smooth medical images after delta coding) are fast-pathed in the decoder to return the cached value without touching the compressed-data slice. Each output symbol in a same run requires only a counter decrement.

D. Large-Alphabet FSE Coding

FSE overview. FSE maintains a single integer *state*. To decode one symbol, the decoder indexes a prebuilt decode table using the current state. Each table entry stores the output symbol, the number of bits n to read from the bitstream, and a base value b for the next state. The decoder reads n bits from the bitstream and computes the next state as $b+$ (bits read). This requires only a table lookup, a bit read, and an addition per symbol; no divisions or multiplications are needed. **Encoding** processes symbols in reverse order (last symbol first), building the compressed bitstream from the end; **decoding** reads bits forward.

The RLE output is entropy-coded using table-based ANS (tANS), implemented as Finite State Entropy. MIC extends FSE to support up to 65,535 distinct symbols. As noted in Section I, one 16-bit code value is reserved as the overflow delimiter, leaving at most 65,535 codes available for residual symbols. This contrasts with the 4,096-symbol limit in Zstandard's FSE implementation, which was designed for byte-oriented streams.

Dynamic table sizing. Encode and decode tables are sized to the actual observed symbol range rather than the theoretical maximum. For 8-bit images stored in 16-bit containers (common in MR), this reduces the working set from 512 KB to approximately 2 KB.

E. Adaptive Table Size Selection

The FSE *tableLog* parameter determines the size of the decode table ($2^{tableLog}$ entries) and the precision of symbol probability estimates. MIC automatically selects tableLog based on the number of active symbols and observation density:

- *symbolLen* = number of distinct symbols observed in the RLE output stream (equivalently, the observed symbol range: max value minus min value plus 1, counting all positions in that range even if some go unused).
- *inputLength* = total number of symbols in the RLE output stream fed to the FSE coder.
- *symbolDensity* $=$ *inputLength*$/$*symbolLen*, i.e., average observations per symbol slot.

Selection rules:

- Default *tableLog* $= 11$.
- Bumped to 12 when *symbolDensity* > 32 and *symbolLen* > 128.
- Bumped to 13 when *symbolLen* > 512 and *symbolDensity* > 16.

Table IV presents the tableLog ablation across selected images.

TABLE IV: TableLog ablation: forced TL=11/12/13 vs. adaptive selection (selected images, Apple M2 Max).

Image	TL=11	TL=12	TL=13	Adaptive
CR	3.47×	3.63×	3.69×	3.69×
MG1	8.00×	8.57×	8.79×	8.79×
MG2	7.98×	8.55×	8.77×	8.77×
MR2	3.14×	3.24×	3.28×	3.28×
RG2	3.85×	4.12×	4.23×	4.23×
RG3	5.64×	5.95×	6.08×	6.08×

Nine images benefit from TL=12 or TL=13 (CR +6.3%, MG1 +9.9%, MG2 +9.9%, MR2 +4.6%, RG2 +9.9%, RG3 +7.8%, XA1 +4.4%, MR3 +0.9%, NM1 +1.9%); the adaptive policy correctly identifies all beneficial cases. Decompression throughput is unaffected by tableLog choice (<5% variation).

IV. Multi-State Entropy Decoding

A. The Serial Dependency Problem

A conventional single-state table-based ANS decoder contains a serial dependency chain:

$$\begin{aligned} &\text{let } e = \texttt{decTable}[s_n], \\ &s_{n+1} = e.\textit{newState} + \texttt{getBits}(e.\textit{nbBits}). \end{aligned} \tag{4}$$

With a table-lookup latency of approximately $L \approx 4$ cycles on modern out-of-order CPUs (L1 cache hit), the loop is limited to roughly one symbol per L cycles regardless of available execution units. Modern CPUs have 4–6 execution ports; a single-state ANS decoder uses exactly one dependency chain, leaving 75–83% of the hardware capacity idle.

B. Interleaved Decoding

MIC breaks this dependency by running multiple independent state machines on interleaved symbol positions. In the four-state design, four independent chains (A, B, C, D) reconstruct positions:

$$\begin{aligned} &\text{chain A} : 0, 4, 8, \ldots \\ &\text{chain B} : 1, 5, 9, \ldots \\ &\text{chain C} : 2, 6, 10, \ldots \\ &\text{chain D} : 3, 7, 11, \ldots \end{aligned}$$

The four chains are completely independent: each has its own state variable, its own sequence of table lookups, and its own bit-extraction operations. The CPU's out-of-order engine sees four simultaneous table-lookup address streams.

Bitstream format. The encoder partitions the output symbol sequence into four interleaved subsequences, encodes each independently (producing four FSE-compressed streams), and concatenates them. The four stream lengths are stored as a 4-entry header so the decoder can locate each stream's start offset. All four chains use the same FSE decode table (built from the joint histogram over all symbols). Initial state values for each chain are stored as part of the per-chain header. Output reordering is performed by the decoder, which writes reconstructed symbols to positions $4k$, $4k+1$, $4k+2$, $4k+3$ from chains A, B, C, D respectively.

Latency model. Let L be the table-lookup latency (cycles). Single-state FSE processes 1 symbol per L cycles. The k-state decoder processes k symbols per L cycles (amortized):

TABLE V: Multi-state decoder latency model and empirical speedup.

States (k)	Amortized latency	Theoretical	Empirical
1	L cy/sym	1.0×	1.0×
2	$L/2$ cy/sym	2.0×	1.3×
4	$L/4$ cy/sym	4.0×	2.0×

The empirical speedup is below theoretical maximum due to: (1) bit-reader refill overhead shared across all chains, (2) instruction cache pressure from the unrolled loop, and (3) memory bandwidth limits on compressed stream reads [28].

C. Correctness and Signaling

Correctness follows from partitioning the output sequence into independent subsequences during encoding and reversing that assignment during decoding; each chain operates on a strict subset of positions with no cross-chain data dependency.

The bitstream signals the decoding mode with a 2-byte magic header: `0xFF,0x02` for two-state; `0xFF,0x04` for four-state; followed by a 4-byte symbol count. In the single-state format, the first byte encodes `tableLog`. Since the maximum supported `tableLog` is 17, a first byte of `0xFF` ($= 255$) is not a valid single-state `tableLog` value and can safely be reused as the extended-format marker. All three formats coexist without ambiguity.

Platform-specific kernels. On AMD64, Bit Manipulation Instruction Set 2 (BMI2) `SHLXQ`/`SHRXQ` instructions provide 3-operand variable shifts without contention on the `CL` register (the x86 shift count register used by traditional variable shifts); on ARM64, `LSLV`/`LSRV` (logical shift left/right variable) serve the same role natively. Pure-Go implementations cover all other targets.

D. Isolated FSE Decompression Throughput

Table VI reports isolated FSE decompression throughput (MB/s over the uncompressed RLE symbol stream) for a representative subset of modalities on both platforms.

*CR single-state is depressed by the `zeroBits` bounds-check path (triggered when any symbol probability exceeds 50%); multi-state interleaving reduces this overhead. The AMD64 BMI2 kernel sidesteps the pure-Go check entirely, yielding +127%.

The 4-state decoder delivers a geometric-mean speedup of +68% on ARM64 and +43% on AMD64 over the 1-state baseline. The smaller AMD64 gain reflects a higher 1-state baseline: the Intel out-of-order engine tolerates the serial dependency chain more efficiently, compressing the relative multi-state improvement. For C and SIMD variant throughput on AMD64, see Table X.

TABLE VI: Isolated FSE decompression throughput (MB/s) for representative modalities. ARM64: Apple M2 Max, pure Go. AMD64: Intel Core Ultra 9 285K, 4-state uses BMI2 assembly (`fse4StateDecompKernel`).

	ARM64 (MB/s)			AMD64 (MB/s)		
Image	1-st	4-st	gain	1-st	4-st	gain
MR	514	841	+64%	600	828	+38%
CT	829	1,295	+56%	941	1,631	+73%
CR	140	966	+590%*	1,994	4,535	+127%
XR	1,444	1,952	+35%	3,195	5,771	+81%
MG1	1,140	1,873	+64%	3,379	4,952	+47%
MG-N	2,944	5,696	+94%	2,722	4,981	+83%
NM1	1,658	2,254	+36%	2,078	2,407	+16%
RG1	1,676	3,128	+87%	2,506	4,336	+73%
Geomean (21)	**1,418**	**2,375**	**+68%**	**2,309**	**3,306**	**+43%**

V. Experimental Methodology

A. Dataset

The evaluation uses 21 de-identified DICOM images spanning 10 modalities, drawn from publicly available datasets: NEMA WG-04 compsamples, NEMA 1997 CD, and the Clunie Breast Tomosynthesis Case 1 (Table VII). All images are de-identified per DICOM PS 3.15 Appendix E; no ethics approval is required.

TABLE VII: Test dataset: 21 clinical DICOM images spanning 10 modalities.

Image	Modality	Dimensions	Raw (MB)
MR	Brain MRI	256×256	0.13
CT	CT scan	512×512	0.50
CR	Comp. radiography	2140×1760	7.18
XR	X-ray	2048×2577	10.1
MG1	Mammography	2457×1996	9.35
MG2	Mammography	2457×1996	9.35
MG3	Mammography	4774×3064	27.3
MG4	Mammography	4096×3328	26.0
CT1	CT scan	512×512	0.50
CT2	CT scan	512×512	0.50
MG-N	Mammography	3064×4664	27.3
MR1	Brain MRI	512×512	0.50
MR2	Brain MRI	1024×1024	2.00
MR3	Brain MRI	512×512	0.50
MR4	Brain MRI	512×512	0.50
NM1	Nuclear med.	256×1024	0.50
RG1	Radiography	1841×1955	6.86
RG2	Radiography	1760×2140	7.18
RG3	Radiography	1760×1760	5.91
SC1	Sec. capture	2048×2487	9.71
XA1	Fluoroscopy	1024×1024	2.00

This dataset spans multiple modalities and image sizes but remains modest for strong generalization claims.

B. Baselines

The current study compares MIC with:

- **HTJ2K** (OpenJPH [29]), via in-process CGO bindings,
- **JPEG-LS** (CharLS [30]), via in-process CGO bindings, and
- **Delta + Zstandard** (level 19).

C. Metrics

The paper reports:

- **Compression ratio**: uncompressed size / compressed size.
- **Decompression throughput**: uncompressed pixel bytes reconstructed per second (MB/s).
- **Encoding throughput**: uncompressed pixel bytes processed per second (MB/s).

D. Benchmark Procedure

All codecs were measured using in-process library calls on the same machine to avoid file-I/O and subprocess overhead. The Go testing framework's `-benchtime=10x` flag was used (10 iterations per benchmark; each iteration processes the full image). Run-to-run variance is <2% on Apple M2 Max and <5% on Intel AMD64.

Platforms and compiler flags:

- Apple M2 Max (12-core ARM64), macOS. Go compiler (gc) with default optimization; CGO-linked C components compiled with `-O3`.
- Intel Core Ultra 9 285K (x86-64/AMD64, mixed P-core/E-core topology), Linux. Go compiler (gc) with default optimization; C components compiled with `-O3 -march=native`.

Codec variants:

- **MIC-Go**: Pure Go, single-state FSE decoder.
- **MIC-4state**: Pure Go, four-state FSE decoder.
- **MIC-4state-C**: C four-state FSE decoder via CGO.
- **MIC-4state-SIMD**: C four-state FSE with BMI2 (AMD64) or scalar (ARM64) via CGO.
- **Wavelet+SIMD**: 5/3 wavelet with blocked column layout and Advanced Vector Extensions 2 (AVX2) kernels (AMD64) or scalar blocked (ARM64).
- **PICS-C-N**: Parallel Image Compressed Strips, a strip-parallel mode that splits the image into N horizontal strips, compresses each strip independently with its own FSE table, and decompresses all strips in parallel using a C pthreads worker pool. Strips are independently decodable.

E. JavaScript and Browser Evaluation

A pure JavaScript decoder (∼20 KB ES module, zero Node Package Manager (npm) dependencies) and a Go WebAssembly build (∼2.5 MB) were implemented. Performance measurements reported in Section VI were obtained in Node.js v24.8 (Apple M2 Max).

VI. Results

A. Compression Ratio

Table VIII presents lossless compression ratios for all codec variants across the 21-image dataset.

Analysis. JPEG-LS achieves the highest lossless ratio on all 21 images, as expected given its context-adaptive MED predictor. MIC's design target is not ratio alone but the ratio-throughput-simplicity tradeoff: on ARM64, MIC-4state-C achieves approximately 91% of JPEG-LS's geometric mean ratio (3.12× vs. 3.44×) while delivering approximately 4× the single-threaded decompression throughput and requiring about 1/5 the source lines.

Mammography achieves the highest MIC ratios (up to 8.79×) due to smooth tissue regions producing near-zero RLE runs.

TABLE VIII: Lossless compression ratios: all codec variants, 21 images. Bold = best ratio per row.

Image	Raw (MB)	Δ+Zstd-19	MIC	Wavelet	PICS-4	PICS-8	HTJ2K	JPEG-LS
MR	0.13	1.95×	2.35×	2.38×	2.28×	2.21×	2.38×	**2.52**×
CT	0.50	2.05×	2.24×	1.67×	2.15×	1.96×	1.77×	**2.68**×
CR	7.18	3.24×	3.69×	3.81×	3.70×	3.71×	3.77×	**3.96**×
XR	10.1	1.43×	1.74×	**1.76**×	1.75×	**1.76**×	1.67×	**1.76**×
MG1	9.35	7.20×	8.79×	8.67×	8.84×	8.87×	8.25×	**8.91**×
MG2	9.35	7.21×	8.77×	8.65×	8.83×	8.85×	8.24×	**8.90**×
MG3	27.3	2.04×	2.24×	2.32×	2.31×	2.34×	2.22×	**2.38**×
MG4	26.0	3.10×	3.47×	3.59×	3.59×	3.62×	3.51×	**3.71**×
CT1	0.50	2.47×	2.79×	2.49×	2.54×	2.29×	2.70×	**3.19**×
CT2	0.50	3.24×	3.49×	2.87×	3.11×	2.72×	3.29×	**4.54**×
MG-N	27.3	2.04×	2.24×	2.32×	2.31×	2.34×	2.23×	**2.38**×
MR1	0.50	1.79×	2.09×	2.14×	2.10×	2.08×	2.13×	**2.30**×
MR2	2.00	2.77×	3.28×	3.34×	3.31×	3.31×	3.35×	**3.52**×
MR3	0.50	3.47×	3.93×	4.09×	3.89×	3.84×	4.33×	**4.51**×
MR4	0.50	3.58×	4.12×	4.18×	4.09×	4.03×	4.21×	**4.49**×
NM1	0.50	4.88×	5.15×	5.02×	5.26×	5.28×	5.76×	**6.28**×
RG1	6.86	1.45×	1.70×	1.70×	1.70×	1.69×	1.63×	**1.72**×
RG2	7.18	3.68×	4.23×	4.32×	4.28×	4.30×	4.32×	**4.51**×
RG3	5.91	5.46×	6.08×	6.82×	6.11×	6.12×	6.99×	**7.31**×
SC1	9.71	3.46×	3.71×	3.70×	3.73×	3.74×	3.85×	**4.73**×
XA1	2.00	4.37×	5.01×	4.94×	5.04×	5.03×	4.88×	**5.39**×

CT with full 16-bit dynamic range achieves 2.24× (MIC) vs. 1.67× (Wavelet): coefficient overflow in the 5/3 lifting step inflates Wavelet compressed size.

PICS strip adaptation. Each PICS strip receives a dedicated FSE table that specializes to its local residual distribution. On large images, this per-strip specialization improves ratio slightly because $\sum_k |S_k| \cdot H(S_k) < |S| \cdot H(S)$ when residual statistics vary across strips.

B. Effect of 16-Bit-Native Residual Coding

MIC outperforms Delta+Zstandard (zstd level 19) by 5–22% on all 21 tested images (geometric mean +14%). The gap is structural: the mutual information $I(X_H; X_L)$ of delta residuals ranges from 0.3 bits/symbol (MR) to 1.1 bits/symbol (MG3), matching the empirical range.

C. Decompression Throughput—ARM64

Table IX presents decompression throughput on Apple M2 Max. MIC-4state-C is the fastest single-threaded decoder on 13/21 images; PICS-C-8 exceeds HTJ2K on all 21 images.

†MR is too small for PICS-C-8; PICS-C-4 is best. MIC-4state-C is 1.7–5.0× faster than JPEG-LS on all 21 images. PICS-C-8 peaks at 3,773 MB/s on MG2.

D. Decompression Throughput—AMD64

Table X presents results on AMD64 where MIC-4state-SIMD activates BMI2/PDEP and Wavelet+SIMD gains AVX2 kernels.

†**MR** is too small for PICS-C-8; PICS-C-4 achieves the highest PICS throughput on this image.

On AMD64, MIC-4state-SIMD leads 16/21 images single-threaded; HTJ2K leads on MG1, MG2, MG4, CT, and RG3. PICS-C-8 exceeds HTJ2K on all 21 images on both platforms.

E. Encoding Throughput

Tables XI and XII report per-image encoding throughput for all codec variants on AMD64 and ARM64 respectively. On ARM64, MIC-4state-C is 2–2.5× faster to encode than pure-Go MIC-Go; JPEG-LS is consistently 2–4× slower than MIC-Go. PICS-8 reaches 1.2–2.1 GB/s on large images, sufficient for real-time PACS ingestion.

F. JavaScript Runtime Results

Table XIII reports Node.js single-threaded performance; Table XIV reports parallel PICS decoding via `worker_threads`.

MG1 (9.35 MB) decompresses in 19.4 ms at 483 MB/s using 12 workers. A web-based DICOM viewer can fetch a `.mic` file directly from object storage and decode it client-side without server-side compute or transcoding latency.

Limitation. These measurements were obtained in Node.js `worker_threads`, which does not fully replicate browser Web Worker performance. Full browser performance benchmarking remains future work.

Practitioner takeaway. For deployments where implementation simplicity is the priority, the pure Go or JavaScript decoder suffices. For maximum single-threaded native throughput on a server, use MIC-4state-C. For large images on multicore systems, PICS-C-8 provides the highest decompression rates on both ARM64 and AMD64.

VII. Discussion

A. Why a Simple Residual Pipeline Can Be Effective

A practical conclusion from the current study is that simple prediction can already remove much of the redundancy in many medical images. When the residual stream becomes concentrated near zero and repeated values are common, a direct residual-domain representation can be highly effective. This assumption is most likely to hold for smooth modalities (MG, MR, CR) and may break down for noisy acquisitions, contrast-enhanced images, high-frequency radiographs with burned-in annotations, or non-natural secondary captures.

Fig. 2 shows the empirical delta-residual distributions for five representative modalities after the avg spatial predictor. All five are sharply concentrated near zero, with MG1 (mammography) placing 69.2% of all residuals at exactly zero, explaining its exceptional compression ratio (8.79×).

TABLE IX: Decompression throughput (MB/s), Apple M2 Max (ARM64). Bold = fastest per row (single-threaded: first 5 cols; PICS: last 3 cols).

Image	MIC-Go	MIC-4s-C	Wav+SIMD	HTJ2K	JPEG-LS	PICS-C-2	PICS-C-4	PICS-C-8
MR	144	**348**	248	265	102	530	**710**†	482
CT	191	**356**	316	307	137	524	955	**1,092**
CR	296	524	**567**	367	153	867	1,635	**2,661**
XR	308	533	**627**	334	108	874	1,666	**3,025**
MG1	482	683	678	**810**	409	1,205	2,112	**3,656**
MG2	479	686	697	**790**	416	1,225	2,120	**3,773**
MG3	308	**531**	422	338	153	864	1,673	**3,117**
MG4	417	**625**	516	548	184	1,093	2,004	**3,689**
CT1	239	**436**	425	362	182	686	1,013	**1,183**
CT2	238	439	**481**	375	175	676	1,041	**1,189**
MG-N	316	**536**	468	340	153	883	1,711	**3,175**
MR1	278	**521**	435	325	116	751	1,207	**1,402**
MR2	333	**563**	498	388	172	913	1,552	**2,466**
MR3	375	**639**	507	441	236	908	1,430	**1,614**
MR4	316	**571**	479	406	197	818	1,341	**1,558**
NM1	327	**632**	575	410	210	888	1,400	**1,679**
RG1	235	406	**584**	332	104	602	1,128	**2,017**
RG2	367	590	**644**	443	193	986	1,803	**3,194**
RG3	374	604	**656**	562	246	1,035	1,944	**3,302**
SC1	375	**587**	388	401	229	1,017	1,861	**3,279**
XA1	331	**576**	459	419	204	928	1,583	**2,493**

TABLE X: Decompression throughput (MB/s), Intel Core Ultra 9 285K (AMD64). Bold = fastest per row within single-threaded and PICS groups.

Image	MIC-Go	MIC-4s-C	MIC-4s-SIMD	Wav+SIMD	HTJ2K	PICS-C-4	PICS-C-8
MR	251	501	**708**	194	570	301	124†
CT	231	403	487	364	**544**	**676**	339
CR	324	599	**744**	723	708	1,738	**2,435**
XR	363	601	**803**	681	570	1,714	**1,994**
MG1	617	789	1,119	746	**1,235**	1,872	**2,514**
MG2	562	800	1,166	848	**1,297**	**2,244**	2,085
MG3	357	633	669	**678**	644	1,823	**2,538**
MG4	523	743	773	**808**	916	2,124	**2,707**
CT1	322	520	**676**	525	657	**857**	423
CT2	293	514	636	**705**	627	**847**	687
MG-N	368	635	669	**705**	643	1,872	**2,191**
MR1	356	609	**766**	532	654	**945**	366
MR2	352	658	**975**	601	749	**1,121**	793
MR3	450	728	**809**	676	802	**920**	705
MR4	390	596	**861**	738	660	493	**701**
NM1	367	**717**	668	715	627	**783**	405
RG1	302	497	593	**705**	557	1,248	**1,494**
RG2	433	676	**861**	811	823	1,841	**1,912**
RG3	460	706	858	881	**894**	**1,969**	1,613
SC1	464	695	**888**	710	728	1,819	**1,889**
XA1	413	693	**836**	538	797	1,173	**1,513**

B. The Cost-of-Complexity Frontier

MIC's core Delta+RLE+FSE pipeline achieves a geometric-mean ratio of 3.12×—within 1% of HTJ2K (3.15×) and 91% of JPEG-LS (3.44×)—while decoding faster than both and fitting in roughly 1,100 lines of Go (∼18× fewer than HTJ2K's ∼20k-line C++ library). JPEG-LS leads on ratio but is the slowest codec evaluated; HTJ2K has no browser decoder.

C. Lightweight Client-Side Decoding

A pure JavaScript decoder (∼20 KB, zero dependencies) and a Go WebAssembly build enable client-side in-browser decoding directly from object storage, eliminating server-side transcoding. Native 16-bit medical image decoding in the browser requires a purpose-built implementation; MIC provides one at a fraction of the footprint of general-purpose alternatives.

D. Pathway to Clinical Impact

MIC could be registered as a DICOM Private Transfer Syntax, allowing PACS vendors to adopt it without modifying the DICOM standard. The ∼20 KB JavaScript decoder enables a direct storage-and-serve distribution model that bypasses server-side transcoding entirely.

VIII. Conclusion

This paper presented MIC, a lossless medical image compression pipeline based on spatial prediction, 16-bit-native run-length representation, and large-alphabet table-based entropy coding. The key results across 21 clinical DICOM images spanning 10 modalities:

- **Compression ratios**: 1.7×–8.9× across the 21 evaluated grayscale images.
- **vs. Delta+Zstandard**: 5–22% better compression on all 21 images (geometric mean +14%), attributable to the structural cost of byte-splitting 16-bit residuals.
- **vs. HTJ2K**: MIC-4state-C exceeds HTJ2K decompression on 19/21 images single-threaded (ARM64) and 16/21 (AMD64); PICS-C-8 exceeds HTJ2K on all 21 images on both platforms.

TABLE XI: Encoding throughput (MB/s), Intel Core Ultra 9 285K (AMD64). Bold = fastest per row.

Image	MIC-Go	MIC-4s-C	MIC-C	Wav+SIMD	HTJ2K	JPEG-LS	PICS-4	PICS-8
MR	121	**290**	273	85	177	71	186	128
CT	180	**359**	311	84	178	104	301	312
CR	233	461	423	120	193	89	732	**1,102**
XR	254	550	519	127	214	95	775	**1,212**
MG1	380	861	820	155	508	239	1,112	**1,651**
MG2	380	857	830	153	507	235	1,119	**1,676**
MG3	256	556	514	97	202	109	832	**1,317**
MG4	336	738	710	107	354	162	1,098	**1,901**
CT1	206	**413**	384	94	216	132	310	329
CT2	192	**371**	340	89	194	132	320	294
MG-N	254	562	529	99	202	107	842	**1,353**
MR1	221	**460**	429	101	195	92	364	347
MR2	275	566	532	107	230	102	643	**857**
MR3	300	**609**	591	119	292	142	498	448
MR4	249	**494**	451	108	226	142	370	368
NM1	237	**467**	444	117	242	132	360	302
RG1	229	485	397	123	198	70	651	**942**
RG2	280	582	488	125	254	112	842	**1,220**
RG3	275	550	512	128	273	143	863	**1,222**
SC1	286	577	551	83	235	169	924	**1,425**
XA1	242	496	471	101	219	114	616	**832**

TABLE XII: Encoding throughput (MB/s), Apple M2 Max (12-core ARM64), all variants. Bold = fastest per row.

Image	MIC-Go	MIC-4s	MIC-4s-C	MIC-C	Wav+SIMD	HTJ2K	JPEG-LS	PICS-2	PICS-4	PICS-8
MR	180	219	243	**305**	102	217	104	111	104	103
CT	208	222	314	332	89	258	166	195	**358**	313
CR	307	312	441	416	138	311	119	488	856	**1,195**
XR	336	322	498	500	126	269	123	411	738	**1,136**
MG1	512	508	621	651	162	676	320	858	1,437	**2,001**
MG2	517	505	644	702	160	700	315	829	1,372	**2,095**
MG3	355	352	469	425	95	293	153	530	862	**1,491**
MG4	458	463	554	503	106	451	234	654	1,186	**1,984**
CT1	292	267	383	388	89	314	204	299	**397**	347
CT2	228	236	356	335	95	294	231	286	**373**	343
MG-N	341	357	424	427	94	309	156	538	929	**1,421**
MR1	309	307	400	**441**	104	271	151	176	301	301
MR2	350	350	498	514	106	339	136	463	672	**821**
MR3	385	379	538	**568**	121	389	183	287	434	516
MR4	305	310	466	**471**	118	354	252	262	363	325
NM1	302	302	438	444	114	346	178	273	323	**474**
RG1	284	315	471	367	127	256	107	349	648	**985**
RG2	390	398	493	520	140	402	155	598	1,034	**1,583**
RG3	374	388	488	485	153	455	198	566	1,029	**1,389**
SC1	414	413	508	466	97	349	297	655	1,170	**1,712**
XA1	335	324	449	463	111	357	154	437	708	**944**

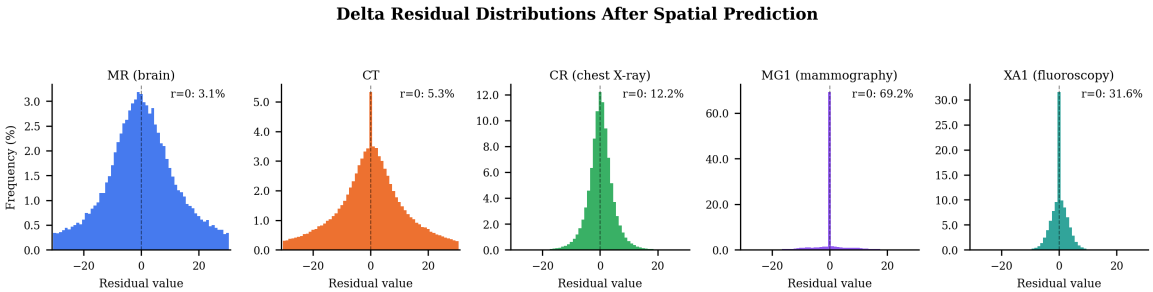

Fig. 2: Delta-residual distributions ($|r| \leq 30$ shown) after avg spatial prediction. MG1's extreme concentration (69.2% at zero) explains its 8.79× compression ratio.

TABLE XIII: JavaScript decoder throughput (4-state), Node.js v24.8, Apple M2 Max. Median over 20 iterations.

Image	Dimensions	Ratio	ms	MB/s
MR	256×256	2.35×	3.0	42
CT	512×512	2.24×	13.5	37
CR	2140×1760	3.69×	161.2	45
MG1	2457×1996	8.79×	80.9	116
MG3	4774×3064	2.29×	638.4	44

TABLE XIV: PICS parallel JavaScript decoder (`worker_threads`), Apple M2 Max. Speedup relative to 1 worker.

Image	Strips	ms	MB/s	Speedup
MR	4	1.3	94	2.57×
CT	4	5.0	101	2.95×
CR	8	31.7	227	5.53×
MG1	8	19.4	483	4.36×

- **vs. JPEG-LS**: 1.7–5.0× faster decompression; JPEG-LS retains the best lossless ratios (geometric mean 3.44× vs. MIC's 3.12×).
- **Multi-state decoder**: geometric-mean +68% isolated FSE speedup on ARM64 (+43% AMD64) with no ratio penalty.
- **Browser decoder**: ∼20 KB pure JS decoder; PICS with `worker_threads` achieves 483 MB/s on MG1 in Node.js.
- **Implementation**: approximately 1,100 lines of Go code, about 18× fewer source lines than HTJ2K.

Overall, the study indicates that 16-bit-native entropy coding is a useful design point for lossless medical image compression, particularly in applications where decompression throughput, implementation simplicity, and deployment portability are important. Future work should strengthen the evidence base with larger datasets, shuffle-based general-purpose baselines, formal statistical reporting, and direct browser benchmarking.

MIC is open-source and available at https://github.com/pappuks/medical-image-codec.

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